Is Stephen Curry really a guard?

Symposium en apprentissage statistique

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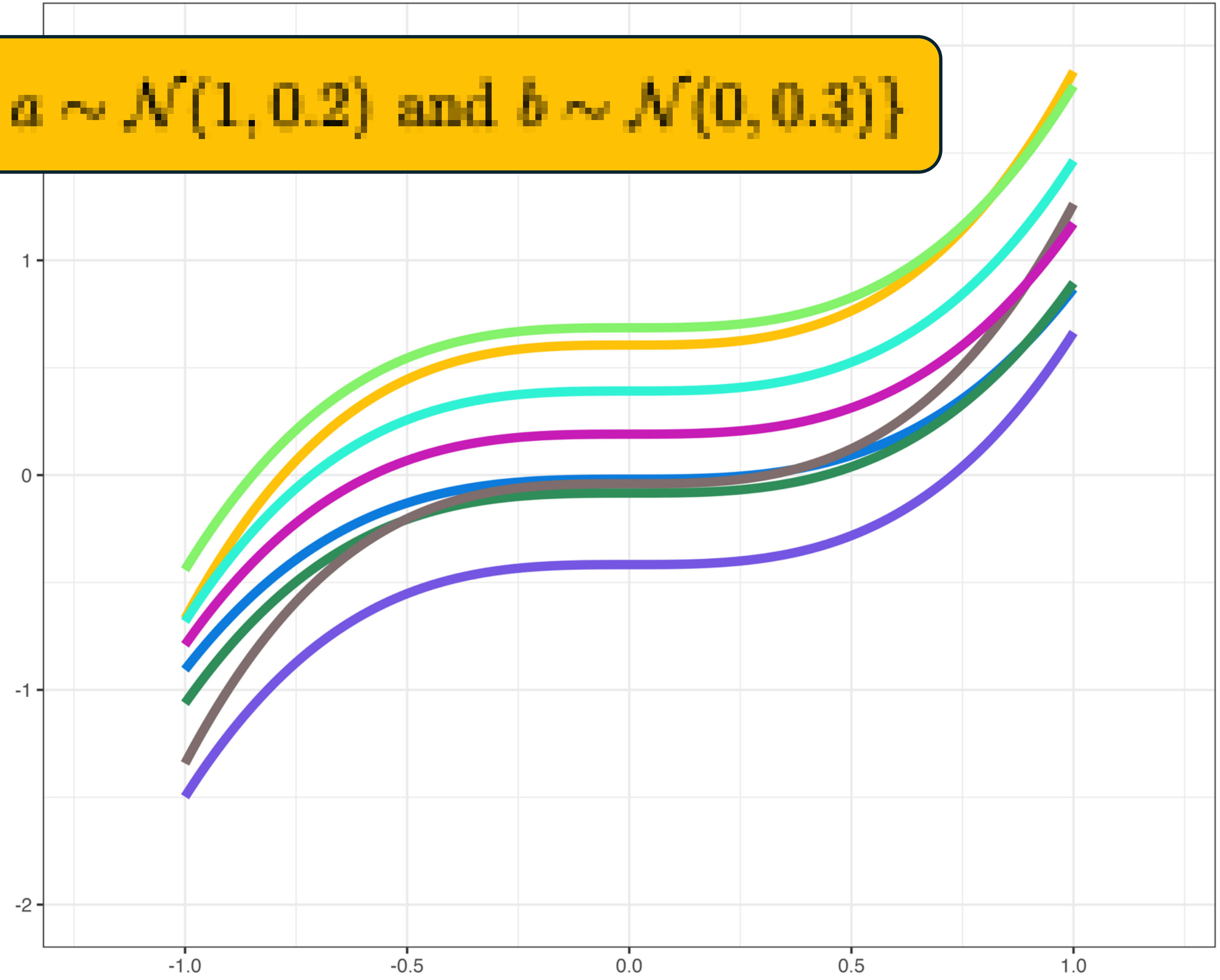
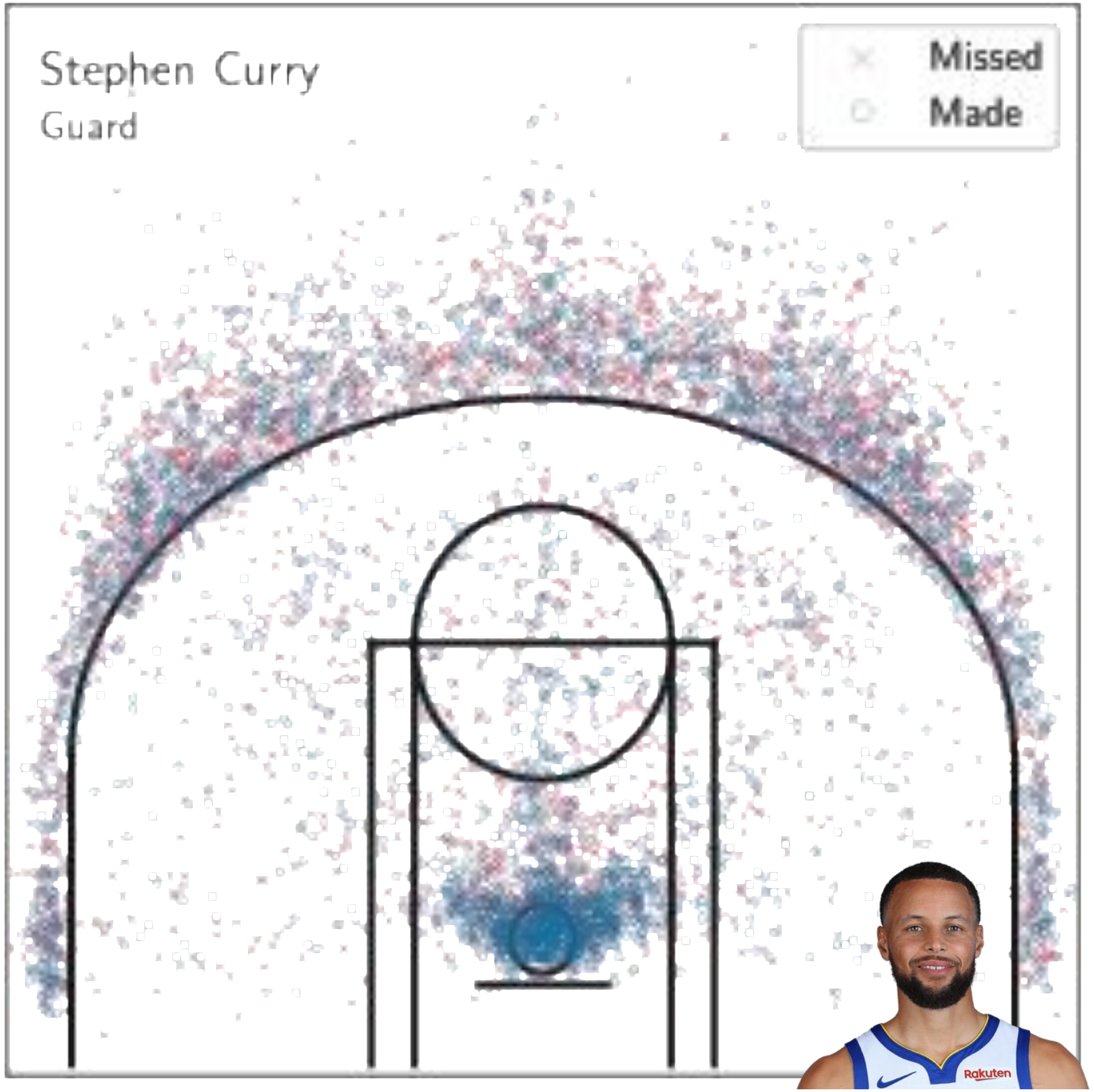
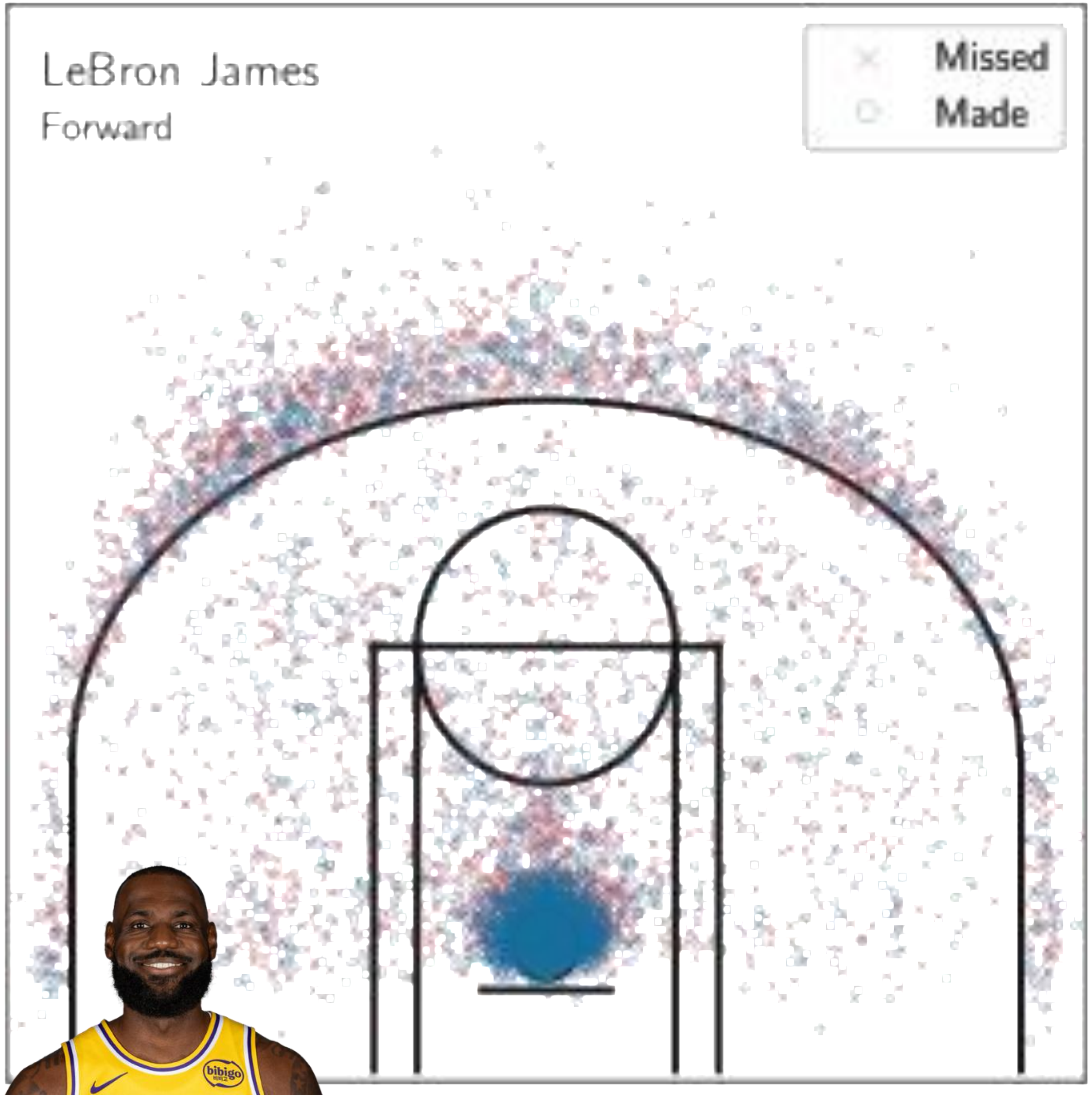
$$\mathcal{X} = \{ax^3 + b, a \sim \mathcal{N}(1, 0.2) \text{ and } b \sim \mathcal{N}(0, 0.3)\}$$



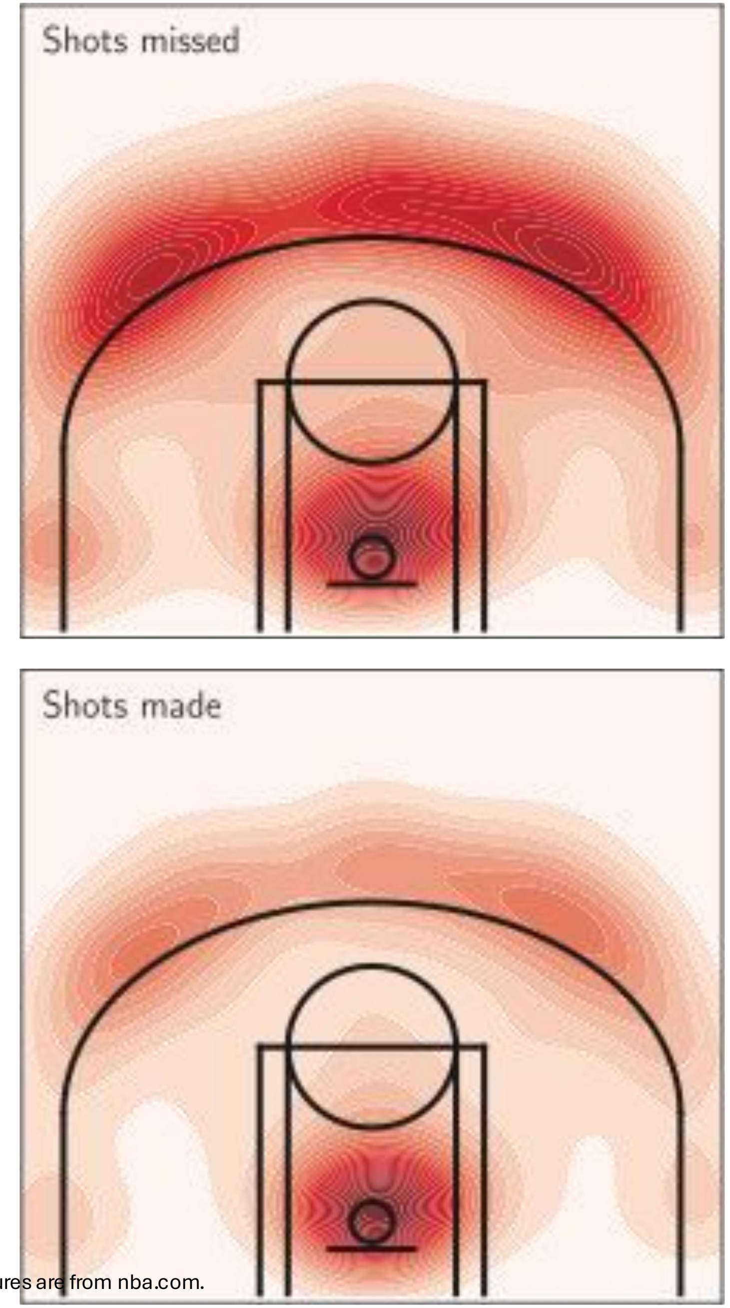


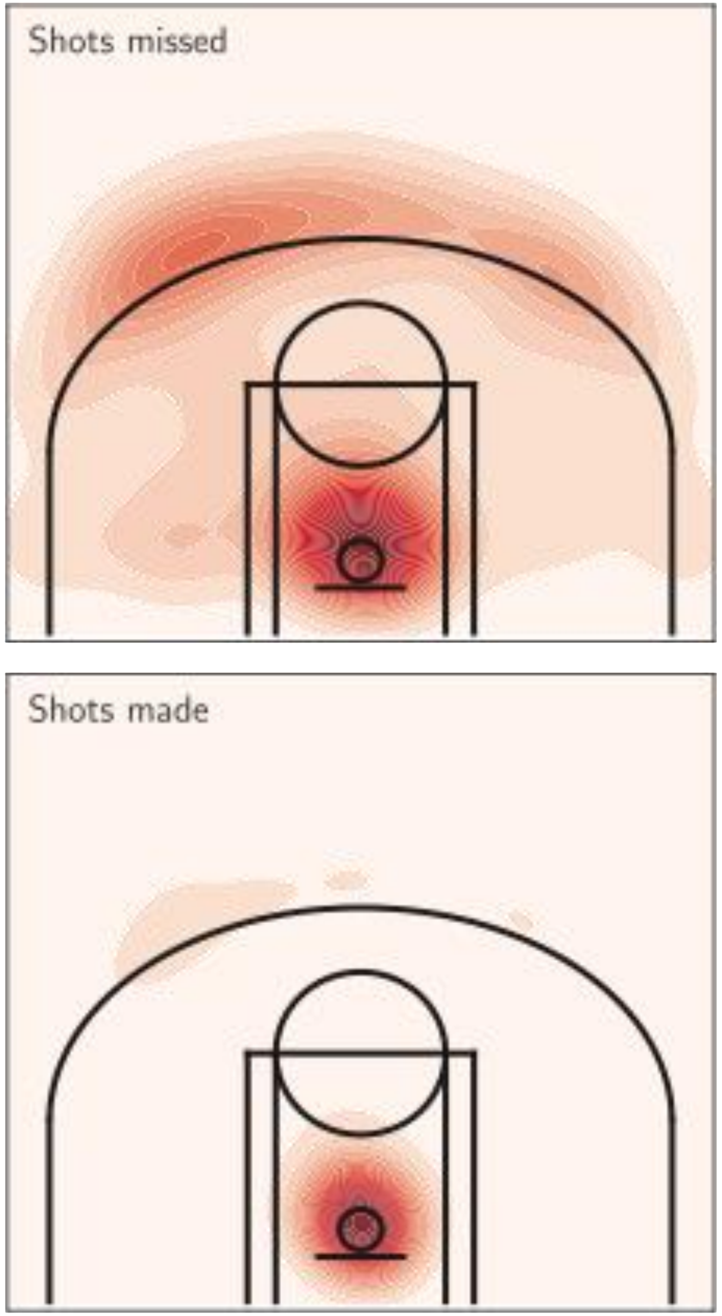


Shots charts are from the authors. Player's pictures are from nba.com.

Stephen Curry - Guard

LeBron James - Forward





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Karhunen-Loève decomposition

$$X(t) = \sum_k a_k \phi_k(t)$$

Solution:

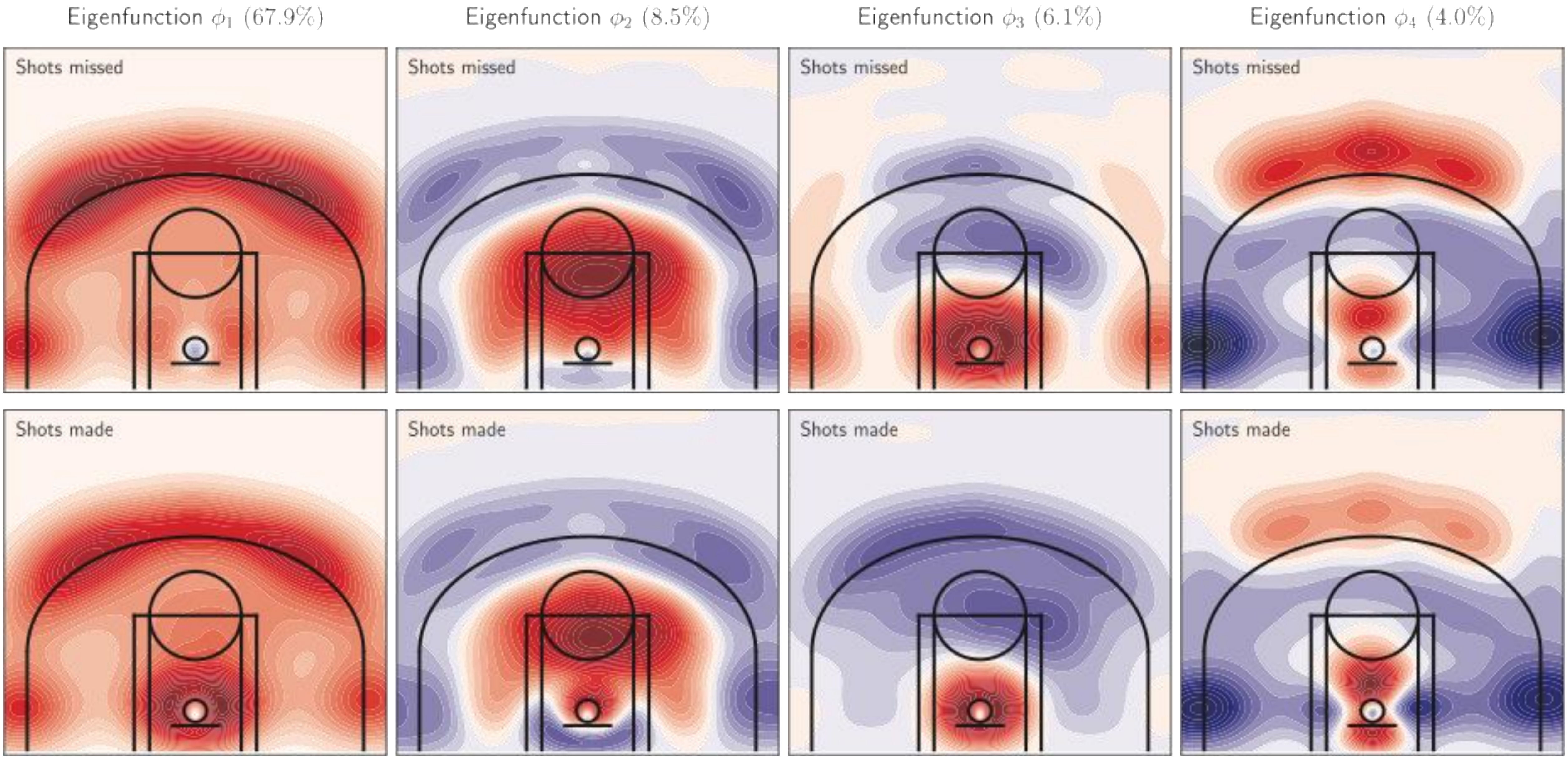
ϕ_k eigenfunctions of the covariance function C

$a_k = \langle X, \phi_k \rangle$

How to compute:

Gram matrix: $M_{ij} = \langle X_i, X_j \rangle$

Golovkine, S. et al., On the use of the Gram matrix for multivariate functional principal components analysis (2025).



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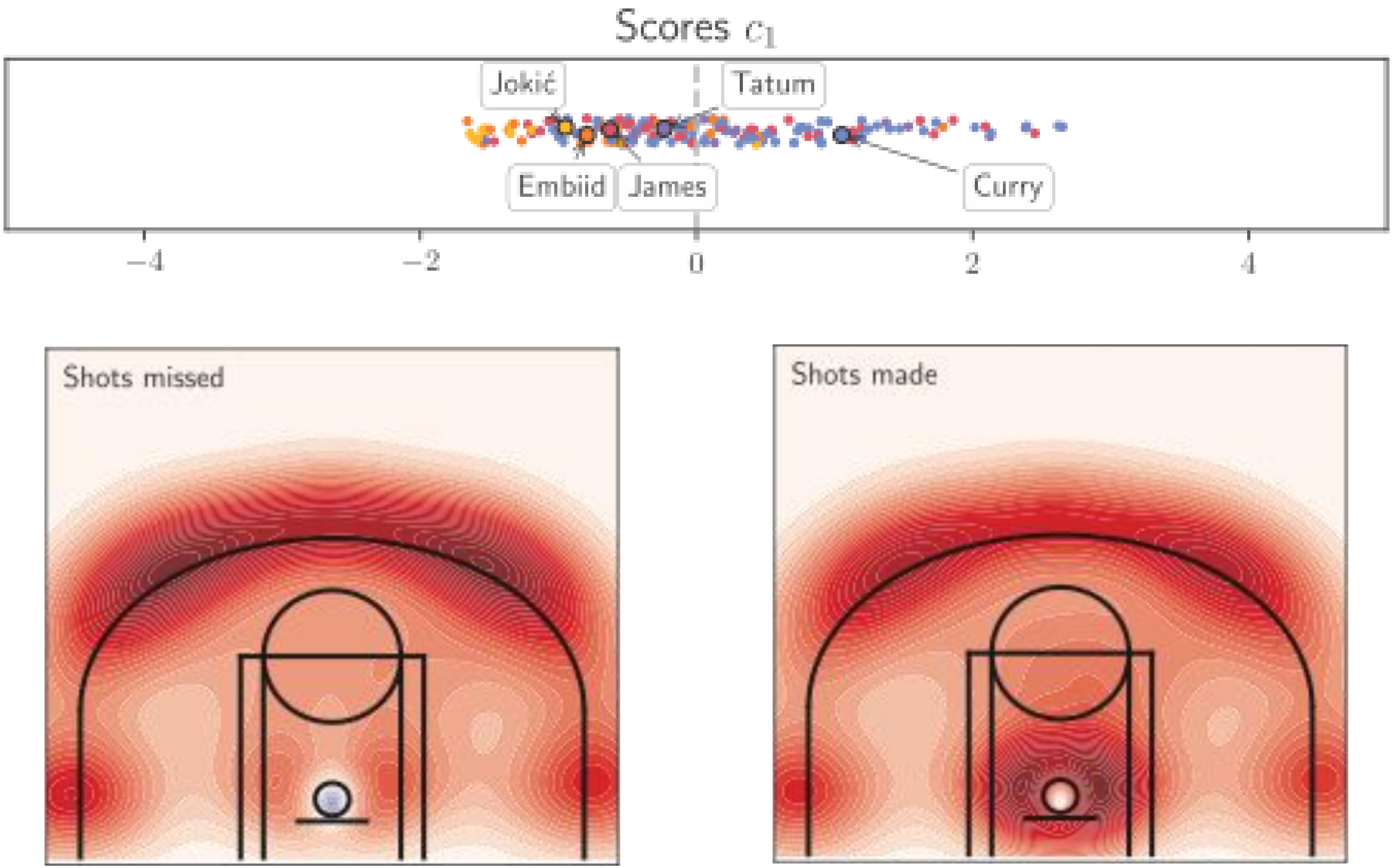

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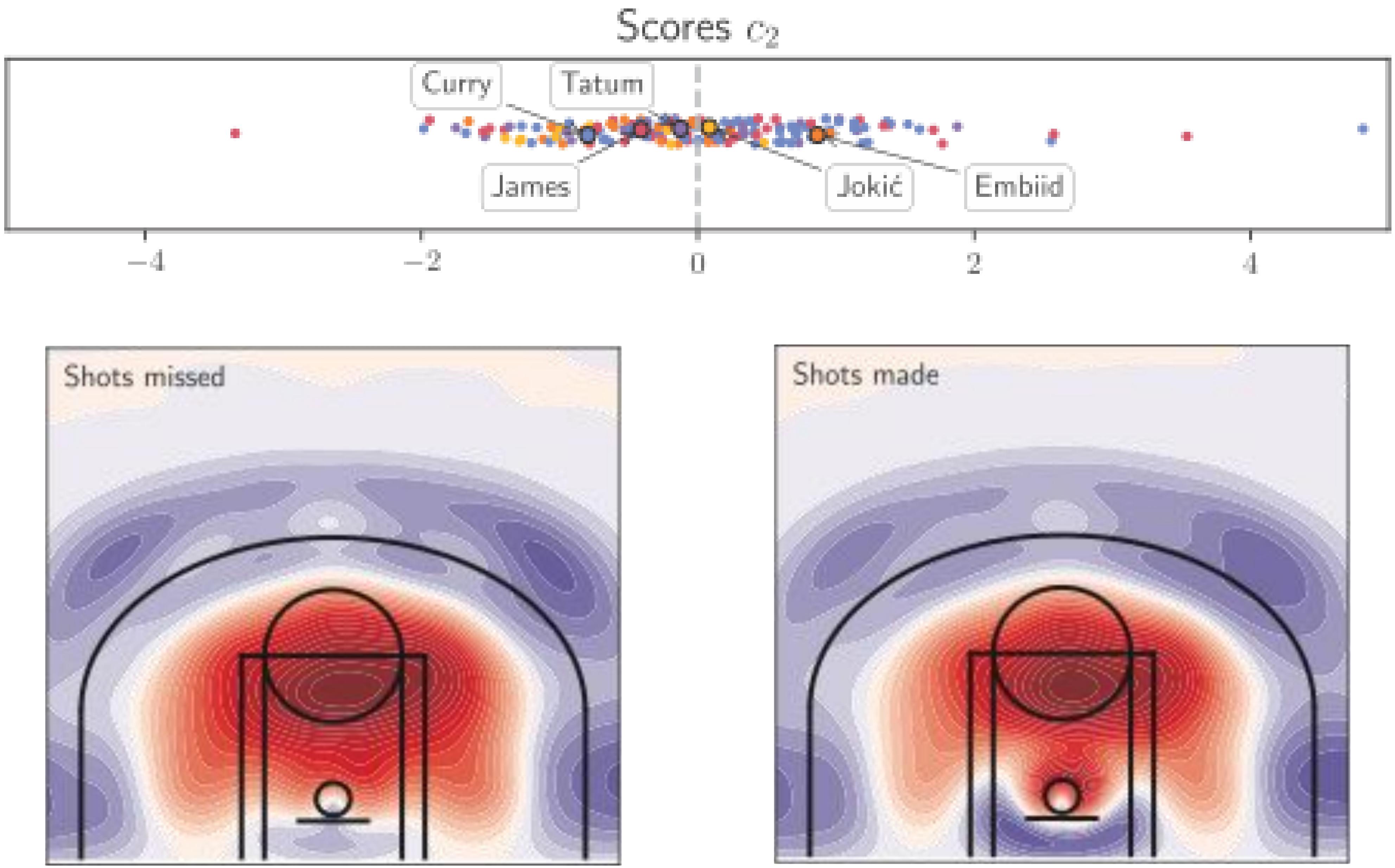

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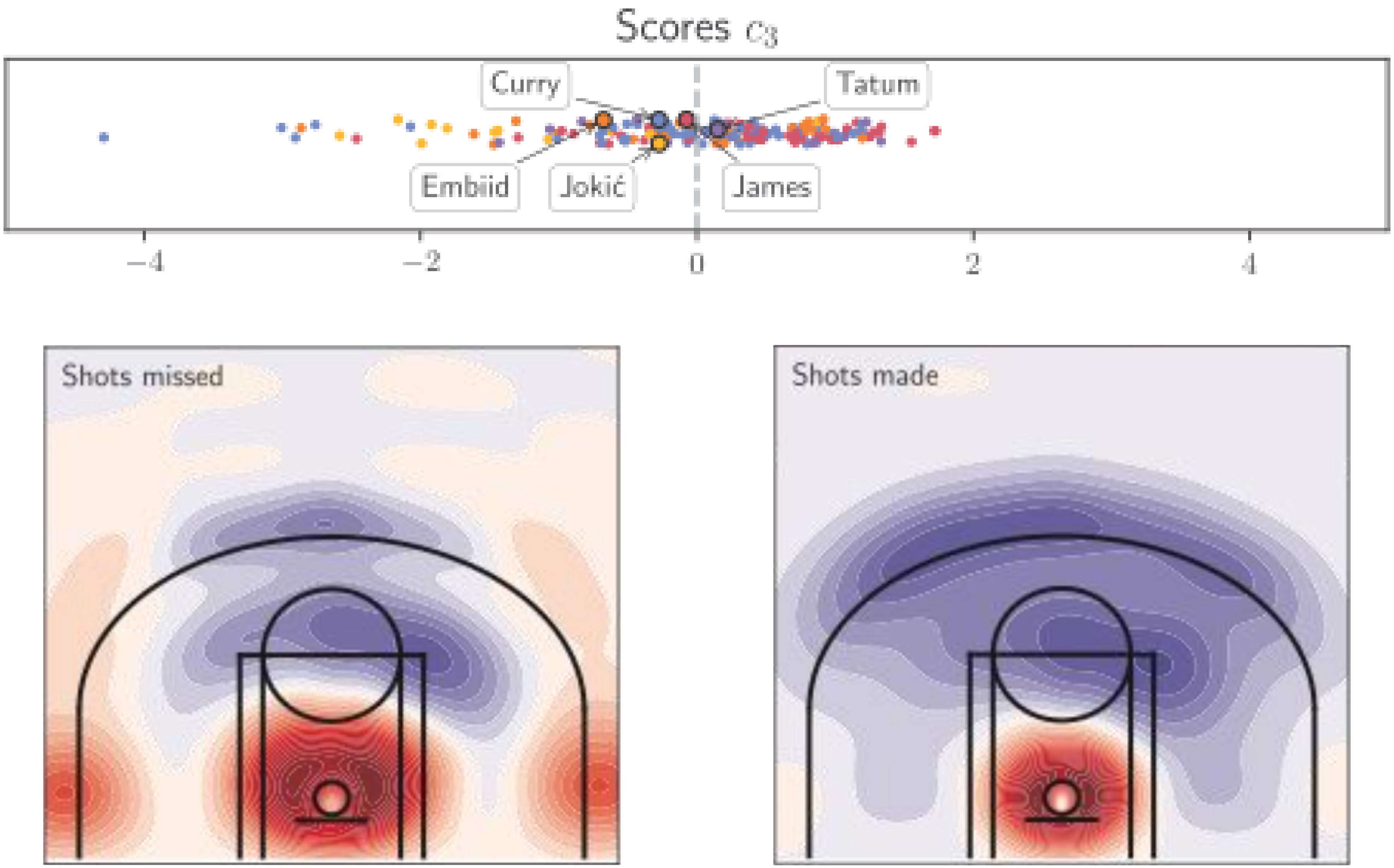

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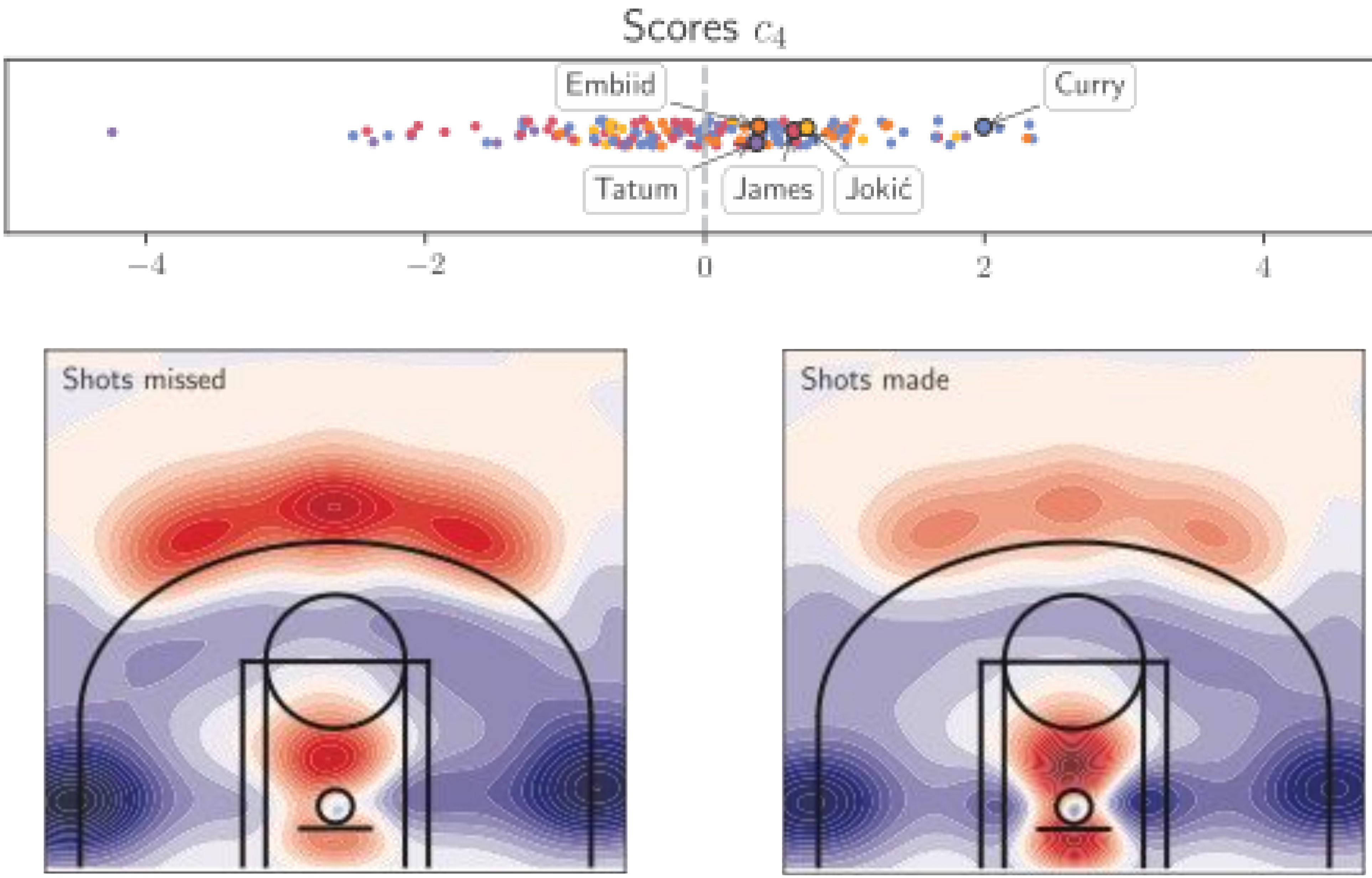

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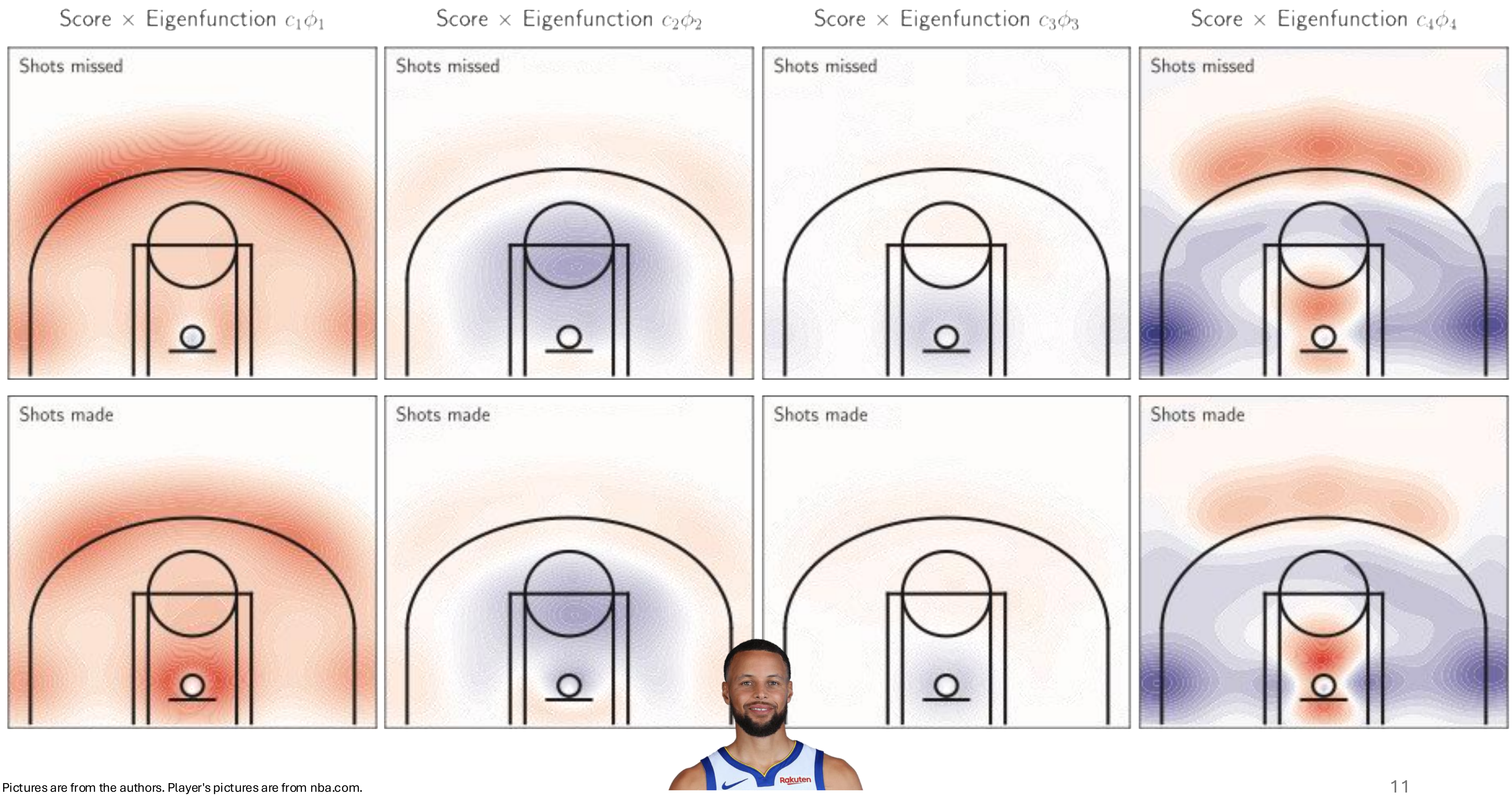

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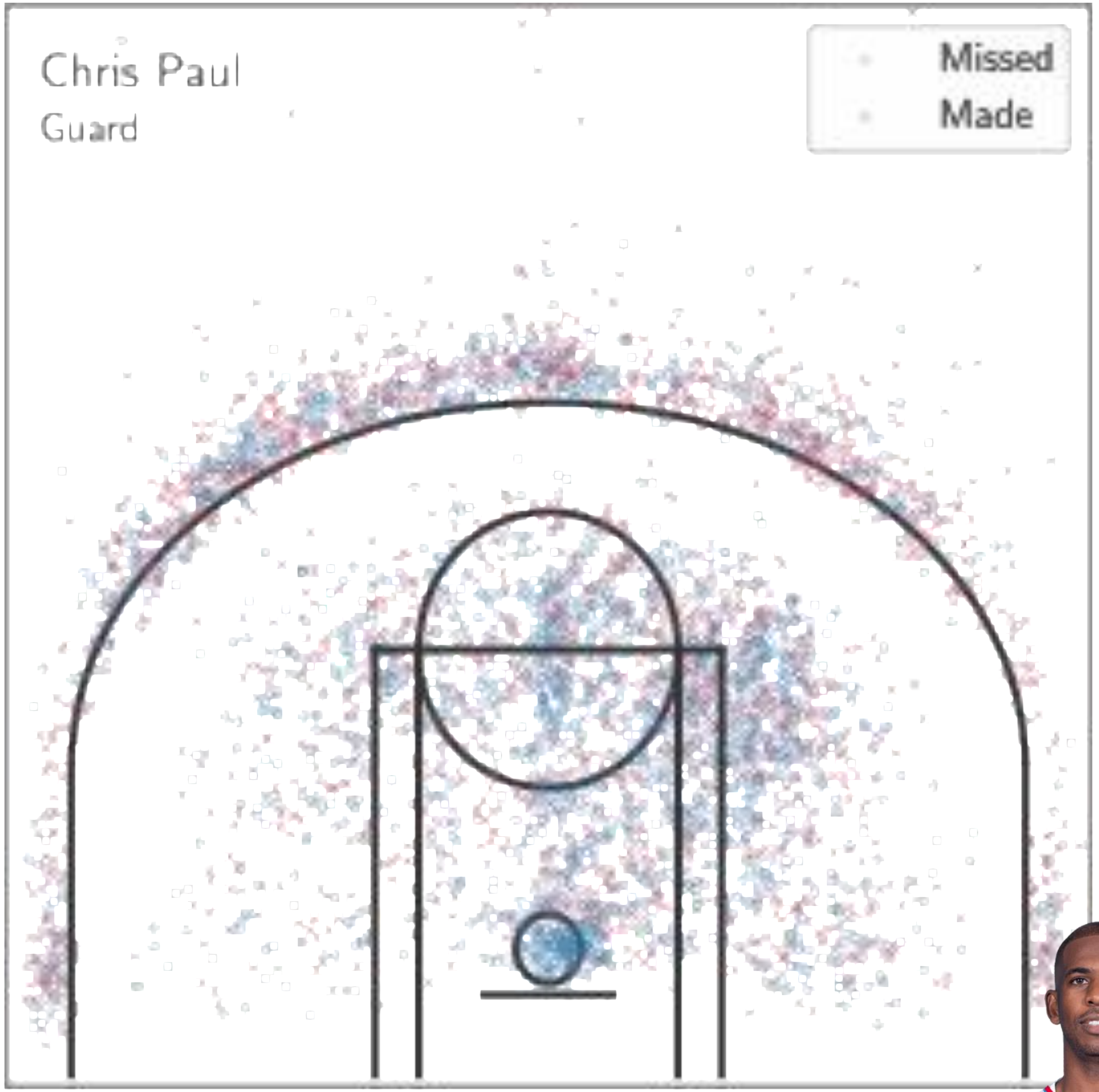

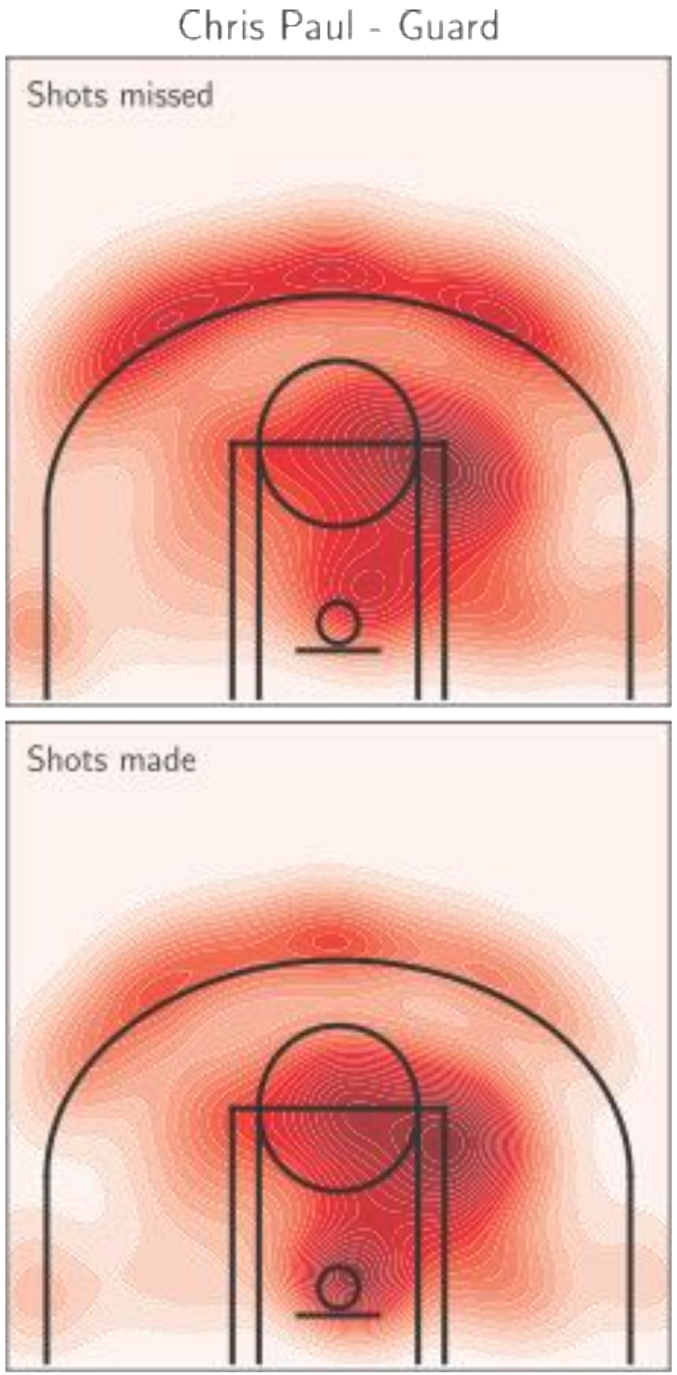

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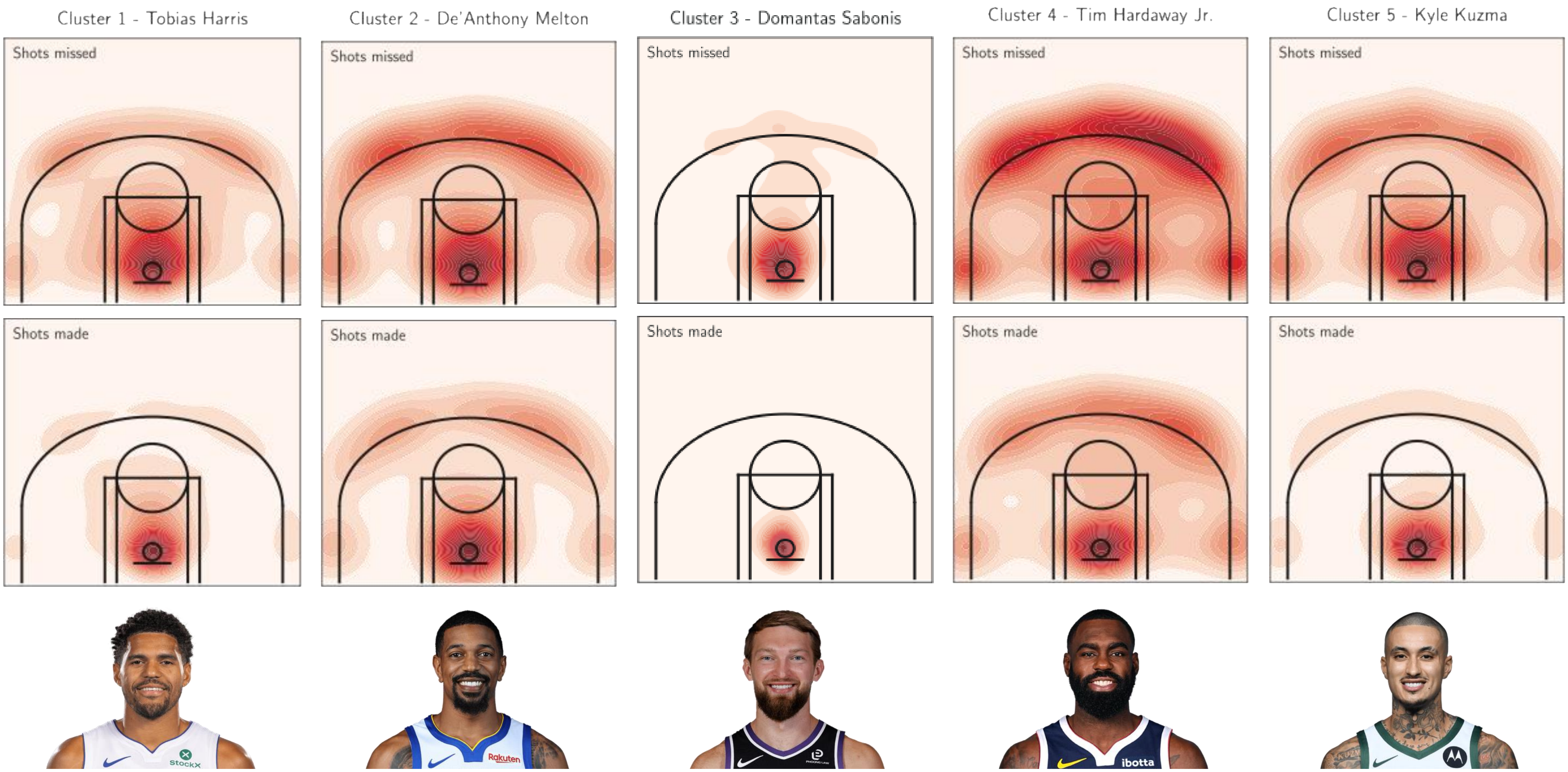
1. Select 5 initial medoids (players).

2. Assign each point to the nearest medoid (player) based on Euclidean distance.

3. Try replacing a medoid with a non-medoid point to see if it reduces clustering cost.

4. Repeat Steps 2 & 3 until no further improvement.



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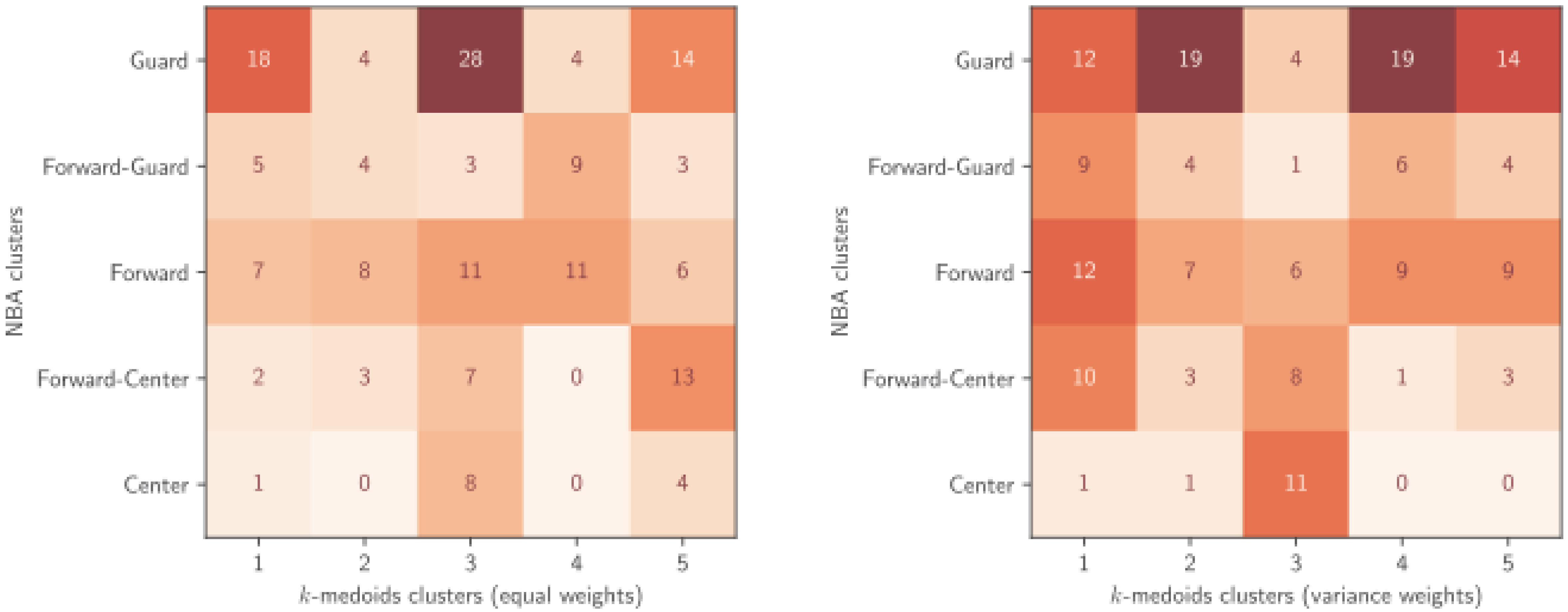

NBA clusters
Guard
Forward-Guard
Forward
Forward-Center
Center
18 4 28 4 14
5 4 3 9 3
7 8 11 11 6
2 3 7 0 13
1 0 8 0 4
1 2 3 4 5
k-medoids clusters (equal weights)
NBA clusters
Guard
Forward-Guard
Forward
Forward-Center
Center
12 19 4 19 14
9 4 1 6 4
12 7 6 9 9
10 3 8 1 3
1 1 11 0 0
1 2 3 4 5
k-medoids clusters (variance weights)

Picture is from si.com.